

Deterministic and probabilistic forecasting of the Lorenz 63 system

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Can the butterfly effect be predicted?

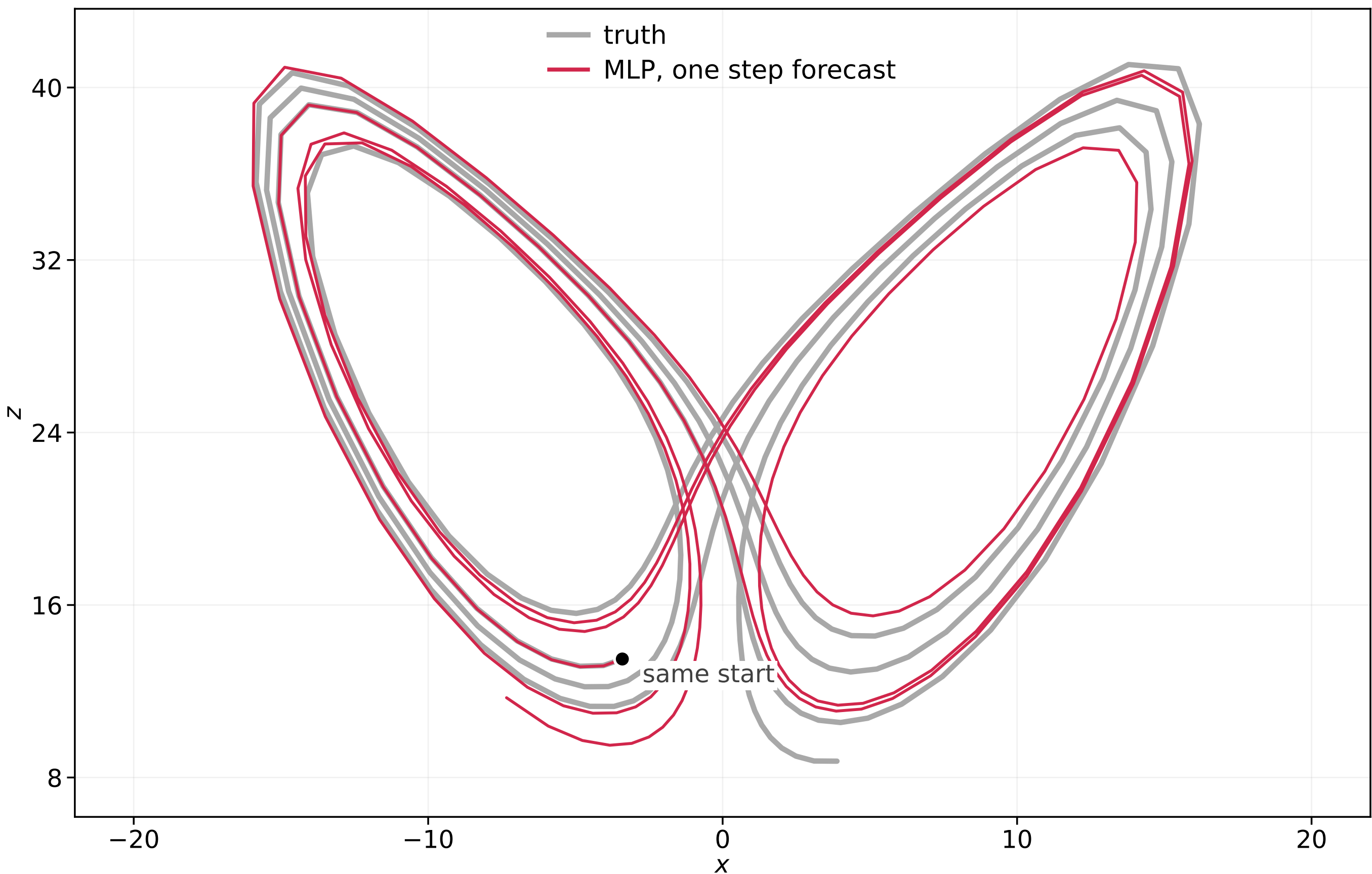
Motivation

Many important systems are chaotic: weather, climate, turbulent fluid flows. A small change in the initial state grows quickly, so individual trajectories over long times are hard to predict **even when the governing equations are known**.

Neural networks forecast far faster than classical solvers, but trained on mean squared error a network learns the **mean** of the next state. This project tests the claim that averaging costs a deterministic model the statistics of a noisy system, and that predicting a **distribution** over next states recovers them.

Objectives

Train five neural forecasters (§2), from a one-step MLP to a Gaussian sampler, on a deterministic and a stochastic Lorenz 63 (§1), under one training budget, and score them all with four measured rulers (§3): **accuracy over short horizons** against **statistical fidelity over long ones**.



1 The system

$\dot{x} = \sigma(y - x)$
 $\dot{y} = x(\rho - z) - y$
 $\dot{z} = xy - \beta z$
 $\sigma = 10 \quad \rho = 28 \quad \beta = 8/3$

Lorenz 63 is the classic minimal chaotic system: a strange attractor, sensitive dependence on initial conditions, and equations known exactly, so every forecast can be judged against the truth. The state is $u = (x, y, z)$; a hat marks a model's prediction. Errors e-fold every $\tau \approx 45$ steps ($\lambda_1 = 0.897 \pm 0.023$, Benettin estimate); one stored step, $\Delta t = 0.025$, is one model call.

Explicit Euler ODE, $b = 0$

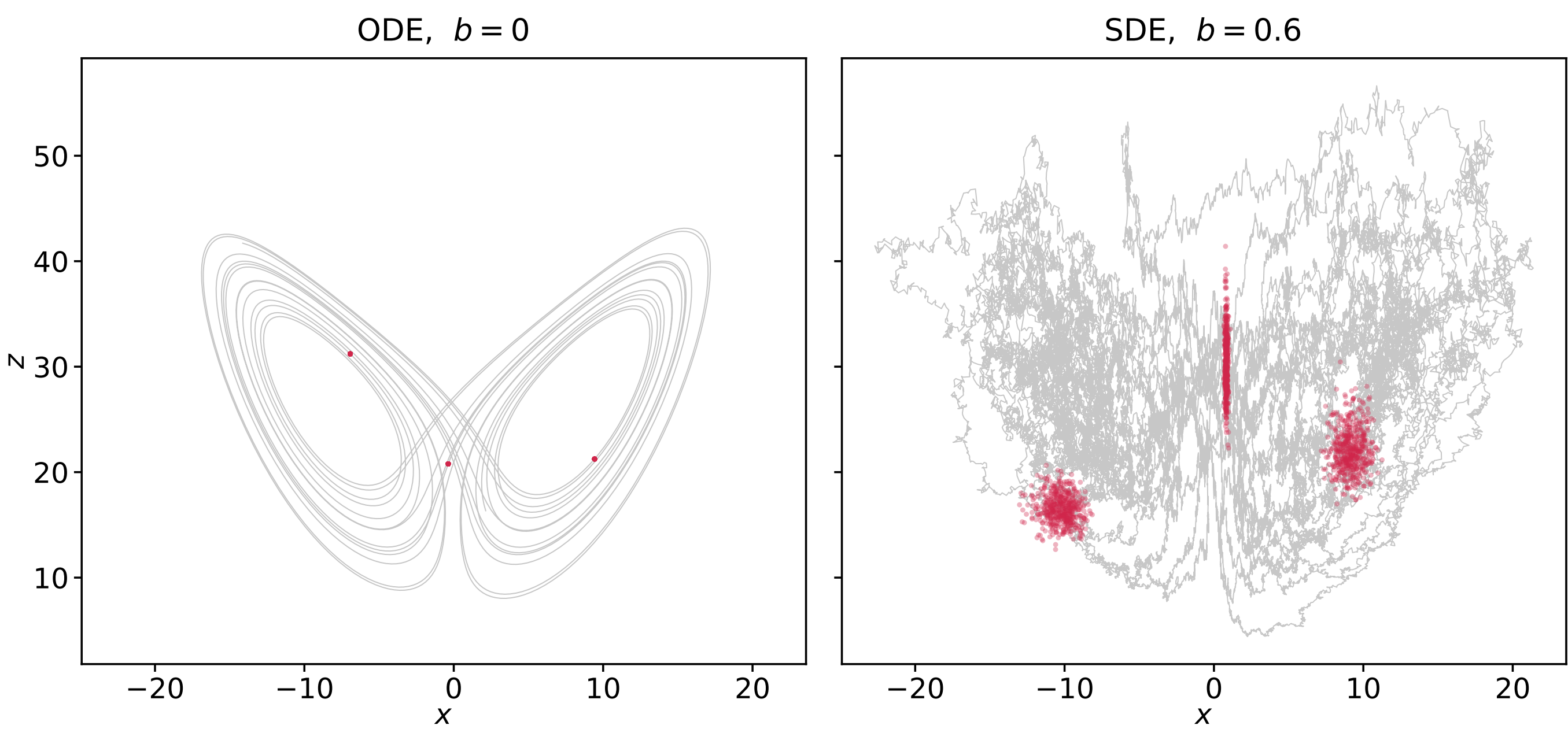
$$u_{n+1} = u_n + h f(u_n)$$

$h = \Delta t / 100$ for the ground truth. At $h = \Delta t$ one step costs the same as **one model call**, the bar every model is measured against.

Euler-Maruyama SDE, $b = 0.6$

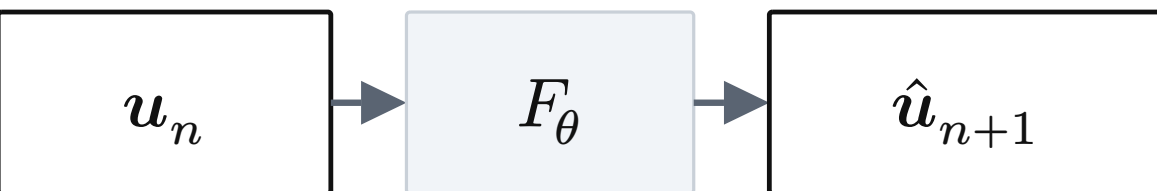
$$u_{n+1} = u_n + h f(u_n) + \sqrt{h} g(u_n) \xi_n$$

The same step and **one term more**, with $\xi_n \sim \mathcal{N}(0, I)$ and $g(u) = bu$, so the noise scales with the state. **b is the project's control parameter**: at 0.6 it spreads the next state over 16.5 % of the attractor scale.

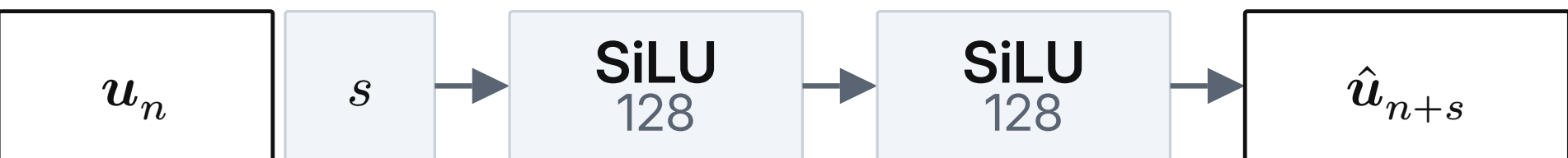


2 The models

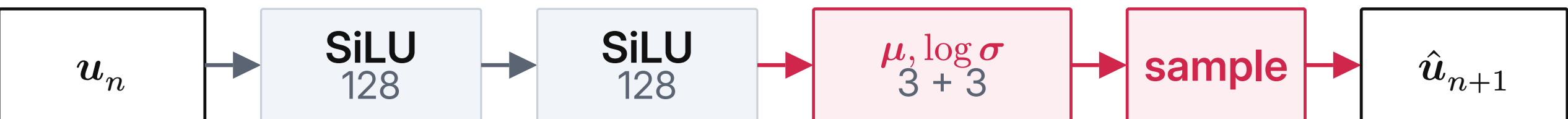
MLP, one step 17 411 parameters · one-step MSE · the baseline



Lead time 17 539 parameters · s = 1...16



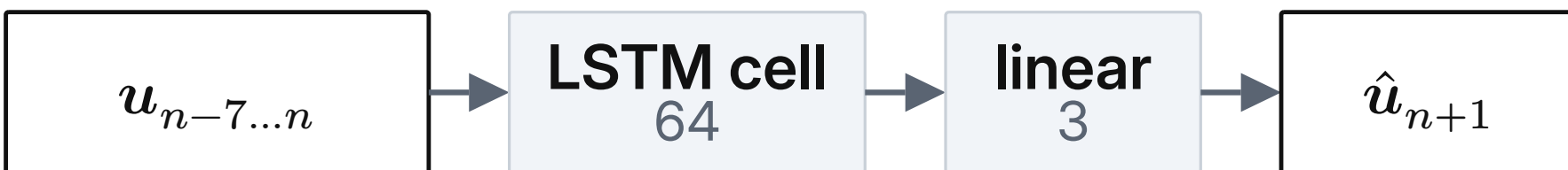
Gaussian 17 798 parameters · diagonal Sigma



MLP, rollout $k = 8$ 17 411 parameters · rollout loss · $k \in \{1, 4, 8, 16\}$ · AR, $k = 8$



LSTM 17 859 parameters · window 8



Every model is **forecast free-running**: handed only a true start (one state; eight for the LSTM), then fed its **own output**, with no correction from the truth. **Red marks what differs from the plain one-step MLP**: the rollout row unrolls the same network's loss k steps **through the composed map**, and the Gaussian predicts a **distribution** and samples it. Widths and depths are held fixed; one trainer runs every model for 3000 Adam iterations on 5 seeds, and one set of rulers judges all five.

3 The rulers

horizon time steps

how long a forecast stays usable

Time steps until the median of $\|u_n - u_n\|$ over 128 starts reaches the attractor scale, the spread of the attractor itself. Past it a forecast is no better than a random state from the attractor.

spread σ ratio

whether it knows what it does not know

Ensemble standard deviation of repeated forecasts from one state, divided by the truth's. 1 means the forecast is as uncertain as the system itself; 0 means the model reports no uncertainty.

climate W_1 ratio

whether long rollouts visit the right places

Wasserstein-1 distance from the model's marginals to the truth's, over the distance between two halves of the truth alone. Marginals only; joint structure is not measured.

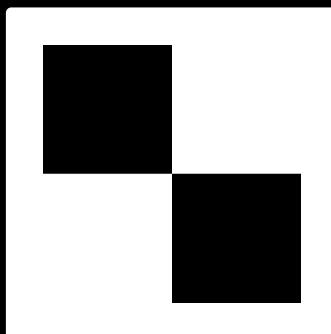
chaos λ_1 ratio

whether it stretches errors like the system

Largest Lyapunov exponent of the learned map over that of the true map, same estimator. 1 means small errors grow at the true rate; 0 or below means the map is not chaotic at all.

4 The results

dot = median over 5 seeds · bar = full seed range · dashed = reference



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